

Alexander Lopez

203-954-8701 | lopez.alexan@northeastern.edu | [linkedin.com/in/lopezalexan/](https://www.linkedin.com/in/lopezalexan/) | New Haven, CT

EDUCATION

Northeastern University, Khoury College of Computer Science

September 2024 - May 2028

- Combined BSE in Computer Science and Mathematics with a Concentration in AI

Cumulative GPA: 3.65/4.0

- **Programming Languages:** C++ | C | Python | Java | JavaScript | OCaml

Major GPA: 3.84/4.0

- **Frameworks and Tools:** PyTorch | TensorFlow | React | SQL | AWS | Pandas | HTML/CSS | Three.js | MATLAB

- **Relevant Courses:** Operating Systems (C, Built an OS from scratch) | Data Structures and Algorithms (Java) | Advanced Probability and Statistics | Number Theory I | Group Theory | Numerical Methods For Differential Equations

Hopkins School

June 2024

- **Extracurriculars:** Math Team | Science Olympiad | Model UN | Diversity Board | Varsity Wrestling (Captain), Football, and Track

EXPERIENCE

Expandya

June 2026 - September 2026

Software Engineering Intern

Westport, CT

- Architected and optimized database systems in Python and C++ to efficiently query and manage datasets exceeding 50M+ rows for eCommerce clients, implementing indexing strategies and query caching that reduced latency by over 1,000% — directly improving site performance and conversion rates under high-concurrency traffic conditions
- Delivered weekly technical presentations to clients, translating complex data insights into actionable eCommerce strategies around site optimization and web performance to drive measurable improvements in customer acquisition and retention

O'Hern Lab at Yale University

June 2023 – September 2023

Computational Biology Research Assistant

New Haven, CT

- Engineered a data pipeline to transform raw 3D protein folding simulations from AlphaFold and NetSurfP neural networks into structured, analyzable datasets; optimized Python scripts on Yale's high-performance computing (HPC) clusters to process 10,000+ cancer-causing protein variants and assess their pathogenic impact based on protein structure
- Developed MATLAB scripts to visualize and interpret simulation outputs, identifying structural patterns across variant datasets; presented weekly progress updates to the research team and delivered a concluding poster presentation to the Yale Department of Computational Biology highlighting key findings in protein-protein interactions (PPI) and related to protein folding

Northeastern University

September 2025 - Present

Teaching Assistant

Boston, MA

- Assist students with coursework for an Intro to Databases course covering SQL querying, data modeling, normalization, indexing, transactions, concurrency, that culminates in a final project of building a full-stack database-backed applications with APIs and dashboards

PROJECTS

Twitter/X Scraper

January 2026

- Built a pipeline to collect and analyze public tweets from any X/Twitter profile without the costly official API, running the data through a three-layer NLP sentiment ensemble combining TextBlob, VADER, and a fine-tuned DistilBERT transformer via HuggingFace and PyTorch to automatically classify each tweet as positive, negative, or neutral

IMC Prosperity 3

March 2025 - April 2025

- Finished top 8% (965 out of 13,000) in a global algorithmic trading competition; working alongside my teammates, researched and designed various trading opportunities that involved market making, scalping, locational arbitrage, and mean-reversion of alpha to optimize both execution and risk-adjusted returns

Momentum Trading Bot

July 2024 – August 2024

- Leveraged Alpaca's API to code a day trading bot that calculates the Relative Strength Index of the Dow Jones 30 and invests in the stock with the lowest RSI if it is less than 20; achieved an initial CAGR of 8.29% and a highest of 20.37%
- Learned about different trading strategies (momentum, mean reversion, and news analysis), order types (bracket, limit, and market), and position management strategies (various stop-loss and take-profit thresholds)

SKILLS AND INTERESTS

Skills: Microsoft Office Suite | Analytical | Communication | Project Management | Problem-Solving | Adaptable | Attention to Detail

Interests: Running (training for my first marathon) | Indoor and Outdoor Rock Climbing | Manchester United (and all things soccer)